DocAI Technical Manual: Security Printing and Physical Anti-Forgery Features

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1. Guilloche Patterns and Iris Printing

Guilloche consists of intricate geometric patterns of fine interlaced continuous lines generated by high-precision mathematical algorithms. Rainbow (iris) printing involves a subtle, continuous colour blend across the substrate without discrete boundary dots. Counterfeit colour copiers inevitably introduce raster moiré dots.

2. Microprinting and Tactile Laser Engraving

Positive and negative microprinting features characters between 0.15 mm and 0.25 mm in height, illegible without magnification (10x). Polycarbonate laser engraving produces localized carbonization inside internal card layers, creating a perceptible tactile relief on the surface that cannot be scraped off without destroying the card.